System Evaluation & Quality Assurance Report

Benchmark Results, ETL Telemetry, and API Performance Metrics

Evaluation Scope: Full Real Clinical Dataset (101,766 Encounters, 71,520 Patients)

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1. ETL Execution & Performance Benchmarks

Metric	Value / Measurement	Evaluation Criteria
Total Encounters Ingested	101,766 Records	Complete dataset volume successfully processed.
Total Unique Patients	71,520 Individuals	De-duplicated patient registry loaded.
Ingestion Execution Time	16.18 seconds	High-speed chunked batch ingestion.
Data Quality Cleanliness Score	100.0% Clean Records	Zero critical identifier or date drops.
Duplicate Detection Rate	0 duplicate encounters accepted	Strict primary key deduplication.
Average Length of Stay (LOS)	4.40 Days	Consistent with published literature.
30-Day Readmission Rate	11.16% (11,357 cases)	Core research outcome metric.
In-Hospital Mortality Rate	1.62% (1,652 cases)	Discharge disposition mapping verified.

2. Automated Test Suite Results (14 / 14 Passed)

Test Suite Module	Tests Passed	Coverage Focus
test_etl.py	3 / 3 Passed (100%)	Missing value cleaning, date parsing, mapping.
test_data_quality.py	3 / 3 Passed (100%)	Duplicate traps, invalid reference codes.
test_database.py	1 / 1 Passed (100%)	3NF schema integrity, foreign keys, cascades.
test_api.py	4 / 4 Passed (100%)	REST endpoints, pagination, search, KPIs.
test_rbac.py	3 / 3 Passed (100%)	Role enforcement (Admin vs Clinician vs Analyst).

3. Summary of Research Findings

Admissions & Readmissions: The overall 30-day readmission rate is 11.16%, with circulatory and diabetes complications driving over 45% of rapid hospital returns.
Length of Stay: Average LOS is 4.40 days, escalating significantly in patients aged 70+ and those admitted via emergency transfer.
Medication Impact: Patients experiencing insulin dosage adjustments ('Up' or 'Down') demonstrate a higher 30-day readmission rate compared to patients on steady insulin regimens, highlighting the importance of post-discharge clinical monitoring.